Bocconi

CORPORATE FINANCE 30017

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Introduction to Portfolio Theory

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Agenda

PORTFOLIO RETURNS

DIVERSIFICATION

THE EFFICIENT FRONTIER

THE CAPITAL MARKET LINE

PORTFOLIO RETURNS

Portfolio Weights

— Share of portfolio total value allocated to specific asset:

$$x_{i,t} = \frac{p_{i,t} n_{i,t}}{\sum_j p_{j,t} n_{j,t}}$$

— where

- p_i is the price of asset i at time t
- n_i is the number of assets i included in portfolio at time t

— A **portfolio can be defined** by its (time invariant) vector of portfolio weights $\boldsymbol{x}$

- *We denote portfolio return by* $r(\boldsymbol{x})$
- *Note: maintaining constant portfolio share requires continuous rebalancing as prices change*

Short Positions

- Negative portfolio weights ($x_{i,t} < 0$)
  - = borrowed assets
  - "Leveraged portfolio" = portfolio with short positions

- Example of short sale:
  1. A borrows 100 shares from B and sells them to C.
     - A gets cash and B receives a cut for the share lease
     - A now effectively owns <u>-100 shares</u> (owing them to B)
  2. A later has to buy shares to return them to B ("close short position")

Portfolio Return

- Portfolio Return is value weighted average of asset returns (under constant portfolio weights):

$$r(\boldsymbol{x}) = \sum_i x_i r_{i,t}$$

- Expected portfolio return is value weighted average of expected asset returns:

$$E[r(\boldsymbol{x})] = \sum_i x_i E[r_{i,t}]$$

- *What time indices do you put if you do not rebalance the portfolio?*

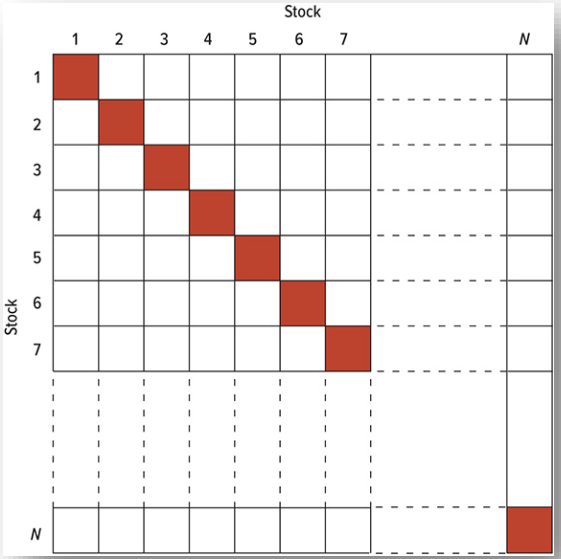
Portfolio Risk: Two Asset Case

$$
\begin{aligned}
\operatorname{var}(x_1\tilde{r}_1 + x_2\tilde{r}_2) &= E\left\{\left[x_1\tilde{r}_1 + x_2\tilde{r}_2 - E\left(x_1\tilde{r}_1 + x_2\tilde{r}_2\right)\right]^2\right\} \\
&= E\left\{\left[x_1\tilde{r}_1 + x_2\tilde{r}_2 - \left(x_1\overline{r}_1 + x_2\overline{r}_2\right)\right]^2\right\} \\
&= E\left\{\left[x_1(\tilde{r}_1 - \overline{r}_1) + x_2(\tilde{r}_2 - \overline{r}_2)\right]^2\right\} \\
&= E\left[x_1^2(\tilde{r}_1 - \overline{r}_1)^2 + x_2^2(\tilde{r}_2 - \overline{r})^2 + 2x_1x_2(\tilde{r}_1 - \overline{r}_1)(\tilde{r}_2 - \overline{r}_2)\right] \\
&= x_1^2 \operatorname{var}(\tilde{r}_1) + x_2^2 \operatorname{var}(\tilde{r}_2) + 2x_1x_2 \operatorname{cov}(\tilde{r}_1, \tilde{r}_2) \\
&= x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2\sigma_{12}
\end{aligned}
$$

	Stock 1	Stock 2
Stock 1	$x_1^2\sigma_1^2$	$x_1x_2\sigma_{12} = x_1x_2\rho_{12}\sigma_1\sigma_2$
Stock 2	$x_1x_2\sigma_{12} = x_1x_2\rho_{12}\sigma_1\sigma_2$	$x_2^2\sigma_2^2$

Portfolio Risk: Multi Asset Case

$$Var\left(\sum_i x_i r_i\right) = \sum_i x_i^2\ \sigma_i^2 + \sum_{i \neq j} x_i x_j \sigma_i \sigma_j \rho_{i,j}$$

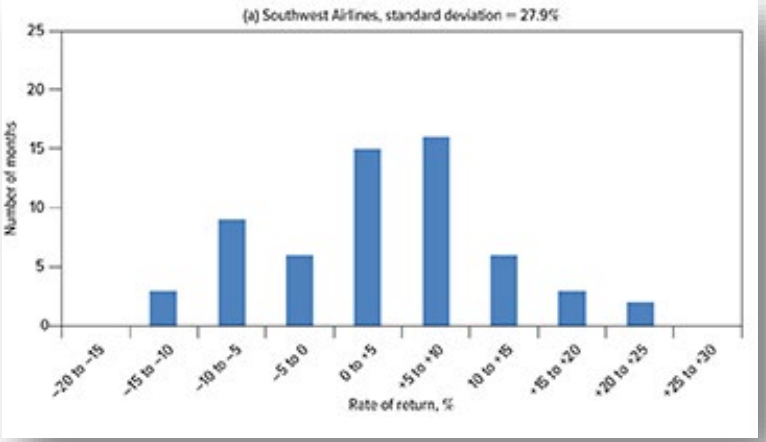


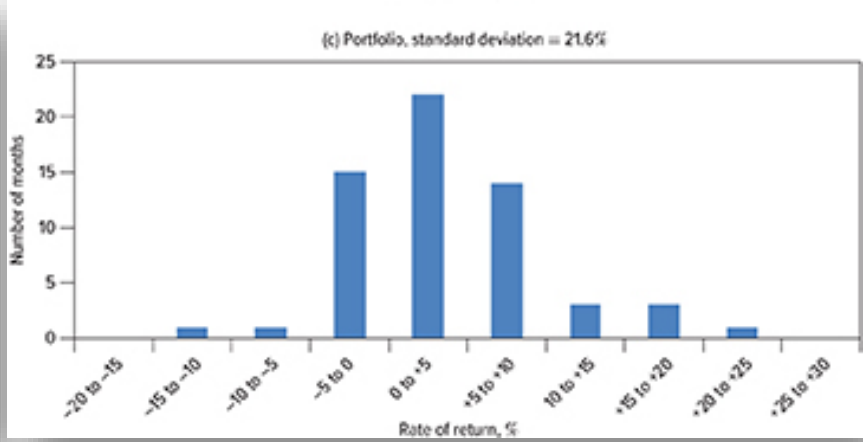
DIVERSIFICATION

Diversification: Intuition

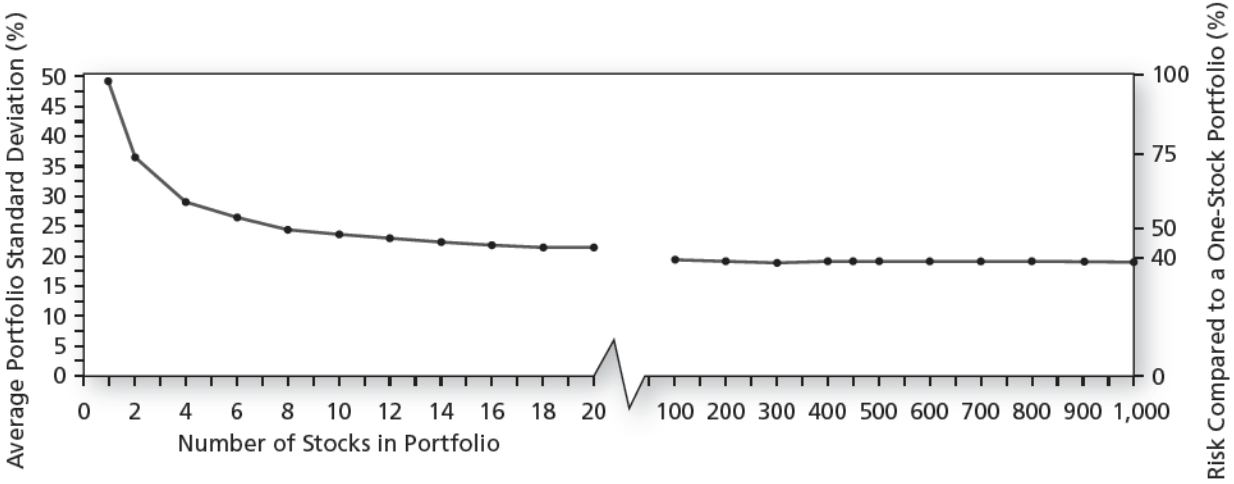
$$Var\left(\sum_i x_i r_i\right) = \sum_i x_i^2\,\sigma_i^2 + \sum_{i \neq j} x_i x_j \sigma_i \sigma_j \rho_{i,j}$$

— *Note: Diversification occurs even with* $\rho = 0$. *Why?*





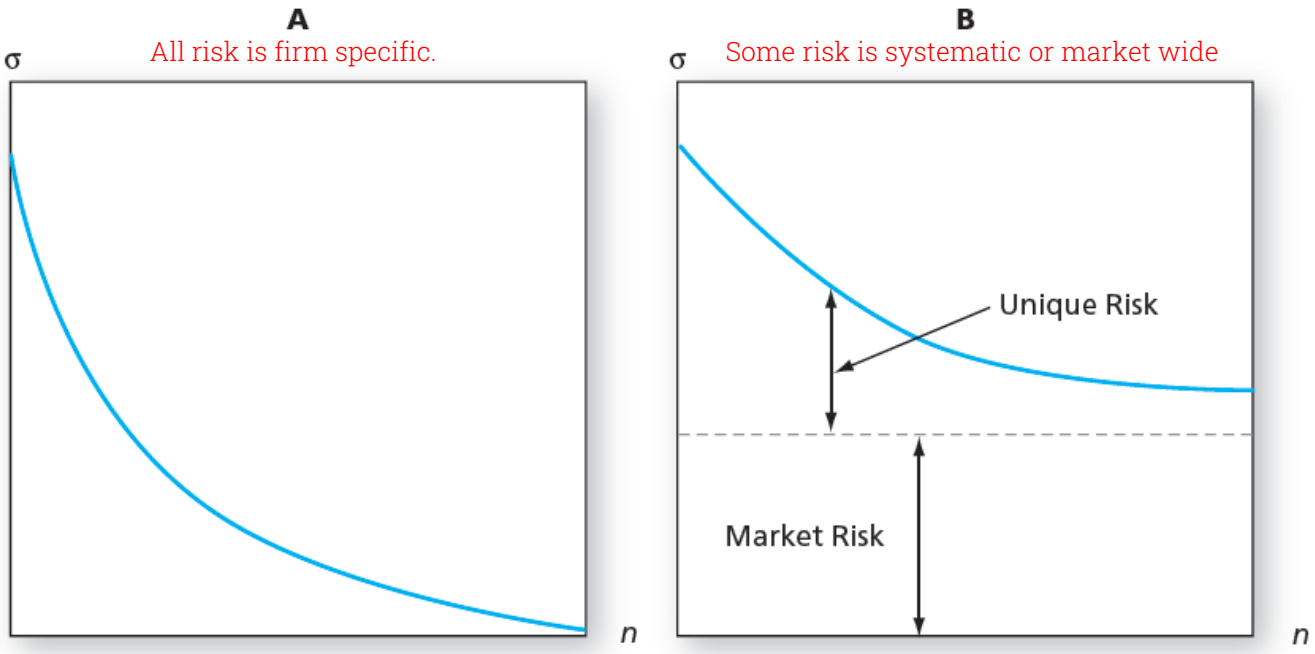
Diversification: Evidence



Elton and Gruber (1977):

- 51 percent of a portfolio SD is eliminated when diversifying from 1 to 10 securities.
- Adding 10 more securities eliminates an additional 5 percent of SD
- Going to 30 securities eliminates an additional 2 percent of SD

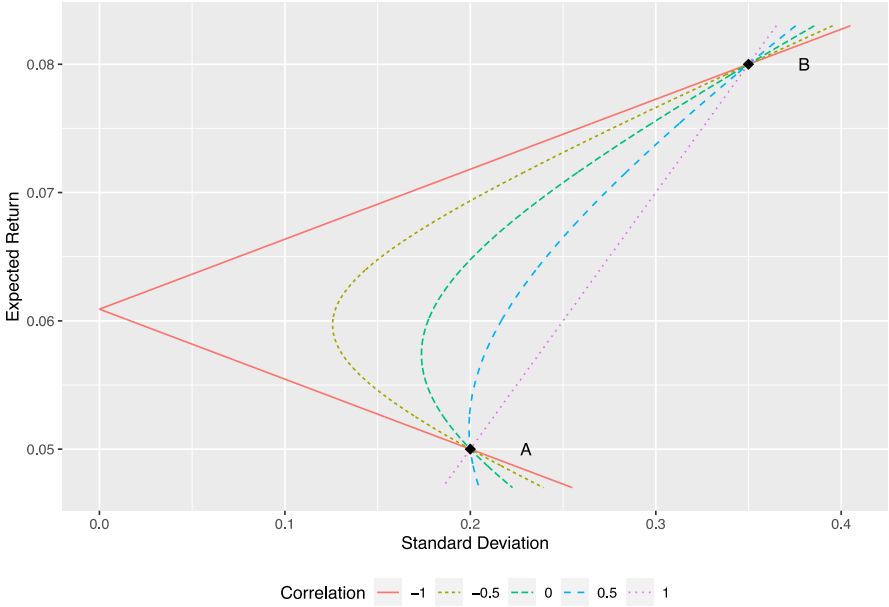
Undiversifiable Risk, aka. *Market Risk*



THE EFFICIENT FRONTIER

Mean-Variance Frontier

- Consider all possible risk-return combinations of a portfolio of two assets (Mean Variance Frontier)
- Consider how Mean Variance Frontier change as assets' correlation changes



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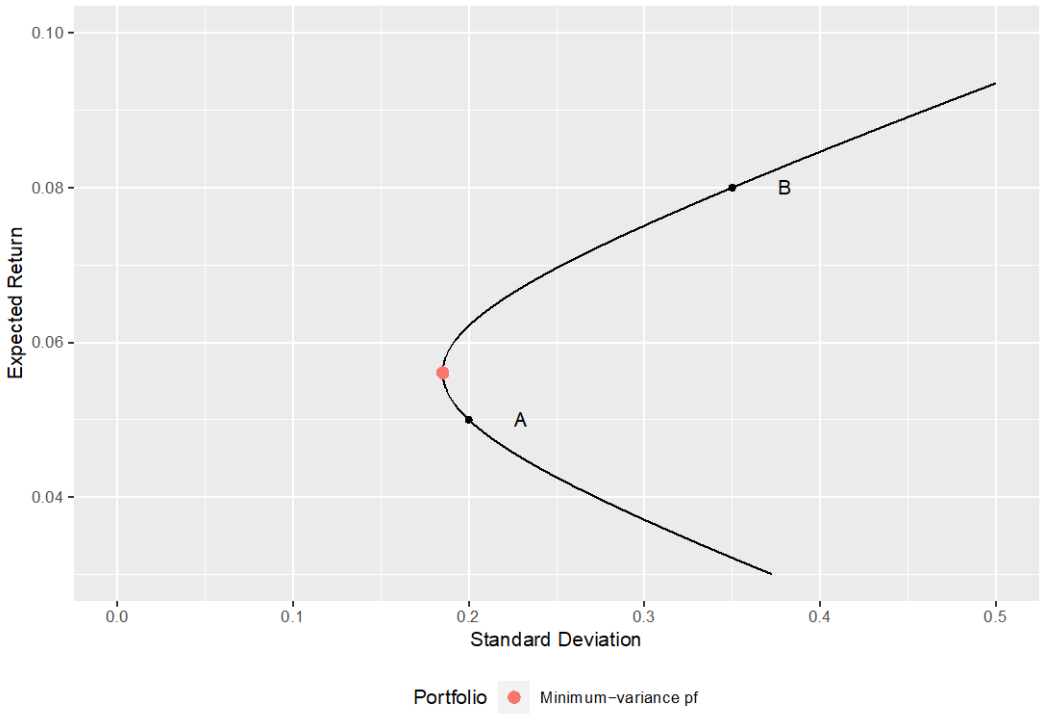
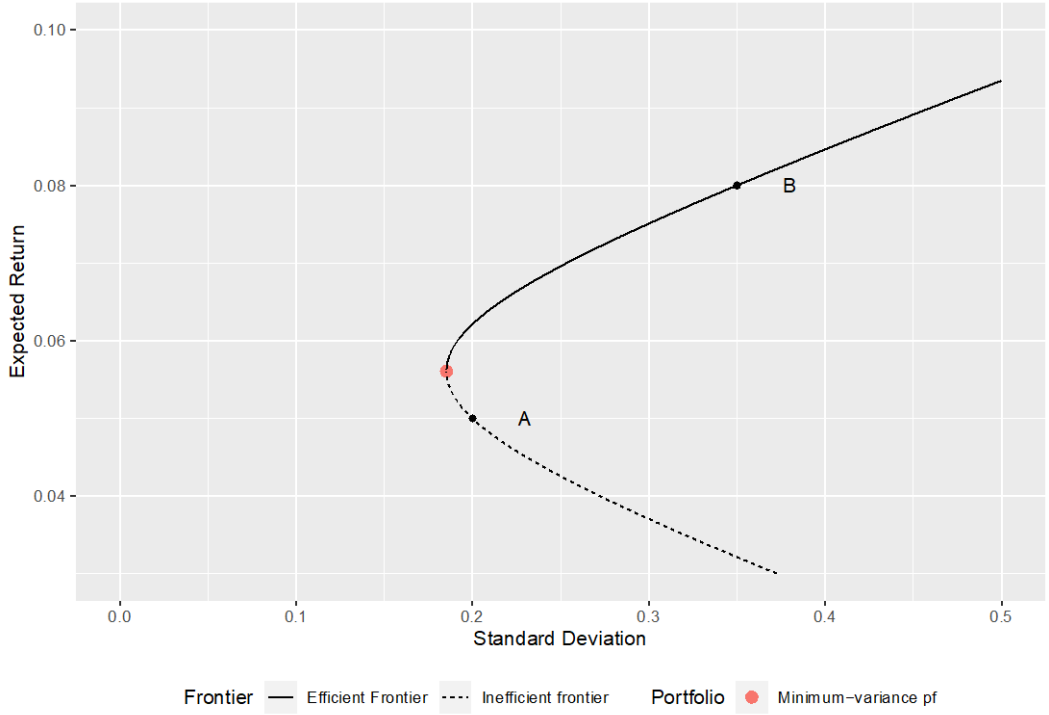
- *What portfolio weight yields zero risk for $\rho = -1$?*

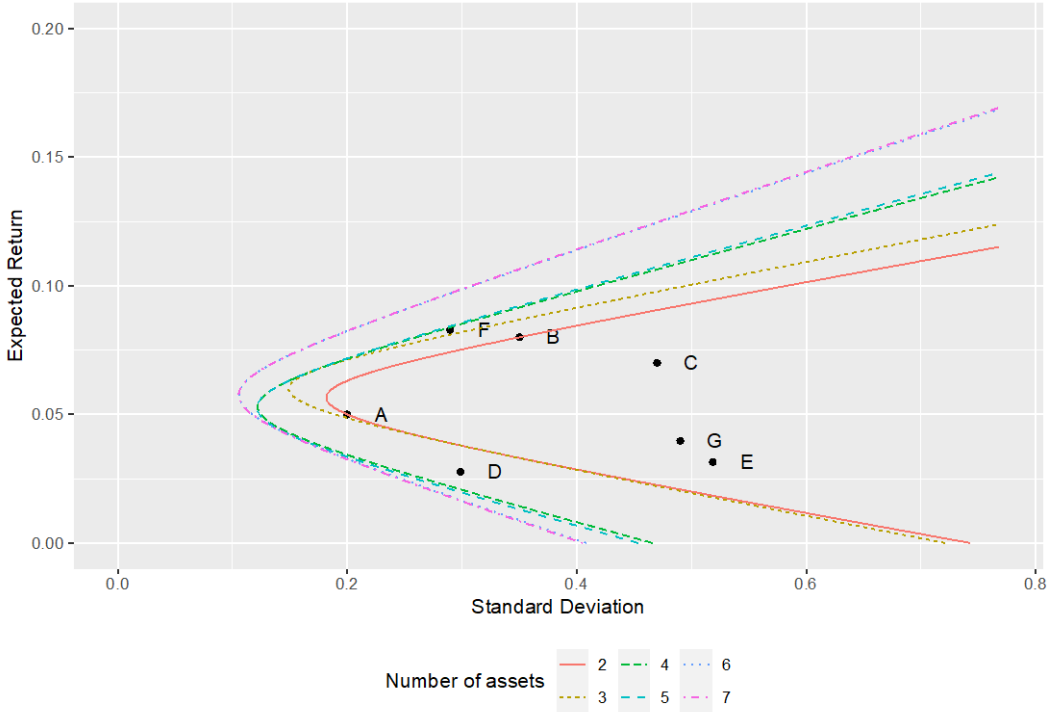
Minimum Variance Portfolio



Reasonable portfolios: *Efficient Frontier*



Diversification in Mean-Variance Diagram



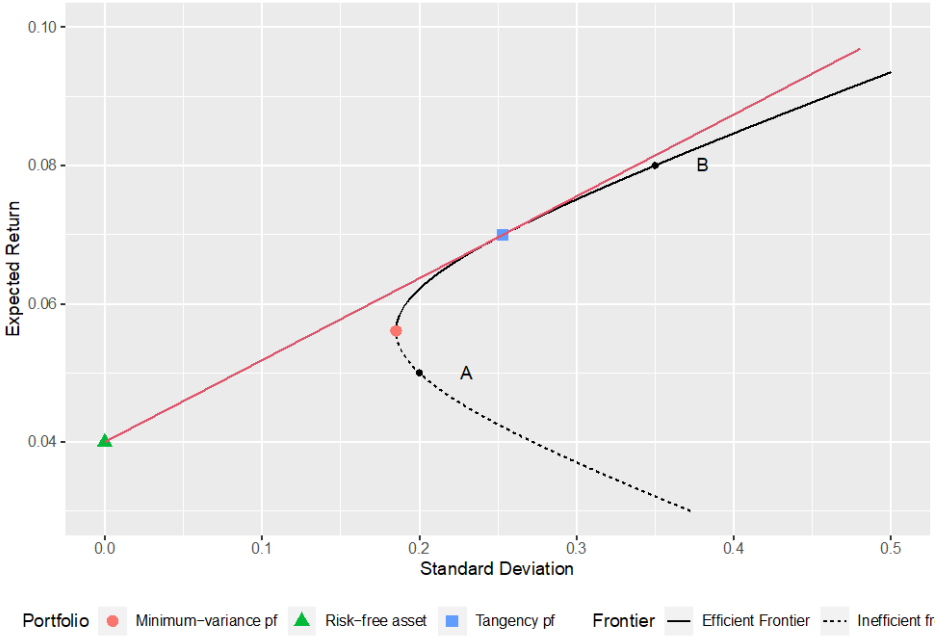
THE CAPITAL MARKET LINE

Adding the Risk Free Asset

- Where would government bonds be located in the mean variance space?

- How does it expand our portfolio's risk-return combinations?
  1. To start, consider two asset portfolio consisting of one risky and the safe asset (*Capital Allocation Line*)
  2. What if the risky asset from step (1) itself was a *portfolio* asset already*?*

The Capital Market Line



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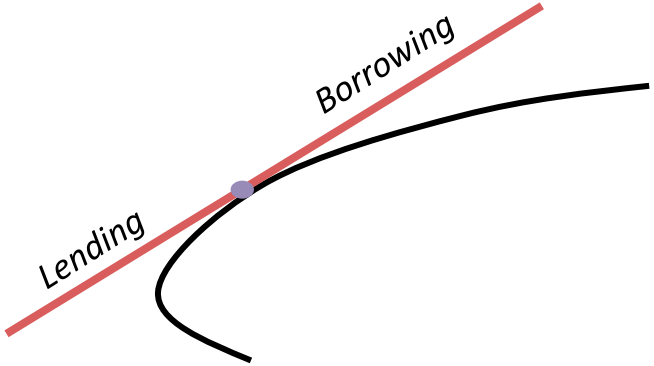
- *What is the difference between Capital Allocation Line and Capital Market Line?*
- *How can we attain points on Capital Market Line above Tangency Portfolio?*

Borrowing and Lending

— *How can we attain points on Capital Market Line above Tangency Portfolio?*

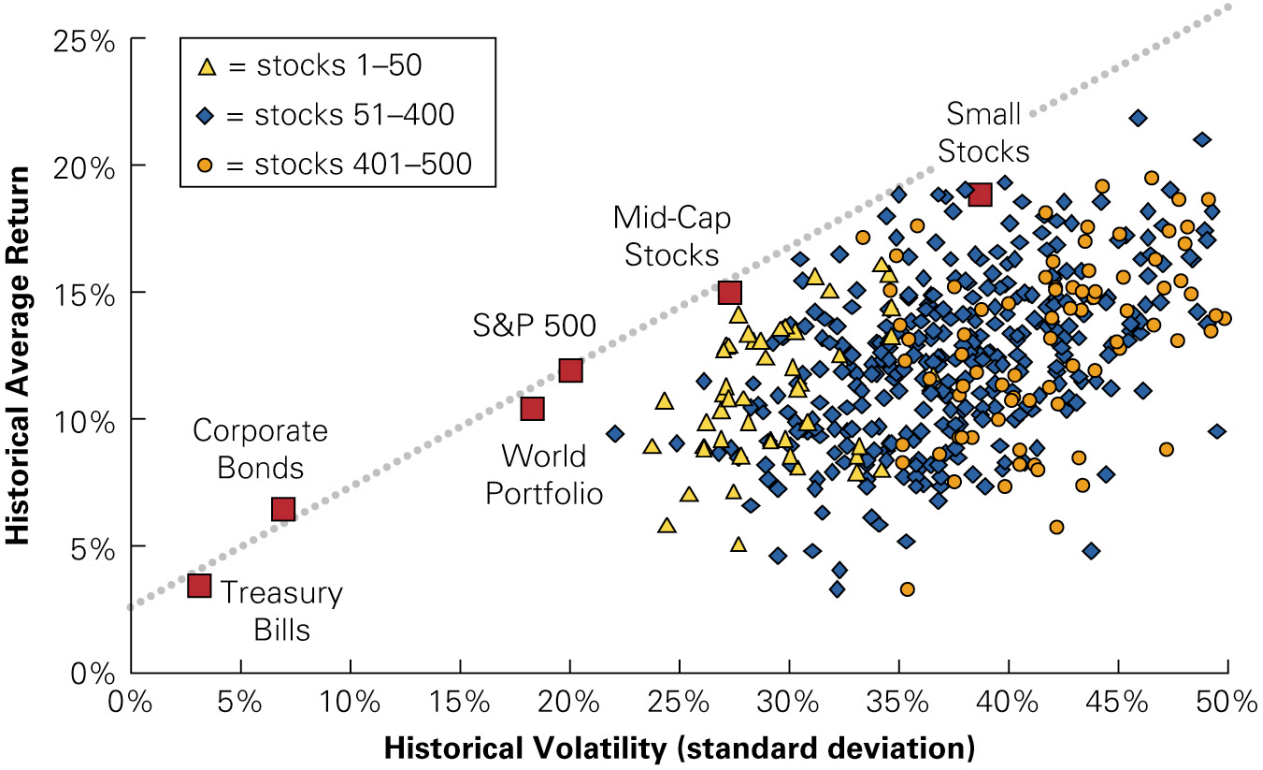
- Short position in safe asset (= borrowing!)



The Sharpe Ratio

- *Sharpe Ratio* of a risky asset captures its risk premium relative to standard deviation of its return:

$$SR_r = \frac{E[r] - r_f}{SD(r)}$$

- **All portfolios on Capital Market Line have the same Sharpe Ratio** (of the Tangency Portfolio)!
  - Portfolios with a lower Sharpe Ratio are inferior and should not be chosen by investors
    - But: does <u>not</u> mean *assets* with lower Sharpe Ratio should be avoided (why?)

Capital Market Line in the Wild